

LECTURE 12

The Entity-Relationship Model

Entities, relationships, and attributes — before a single table is created

CSC270 · Database Systems

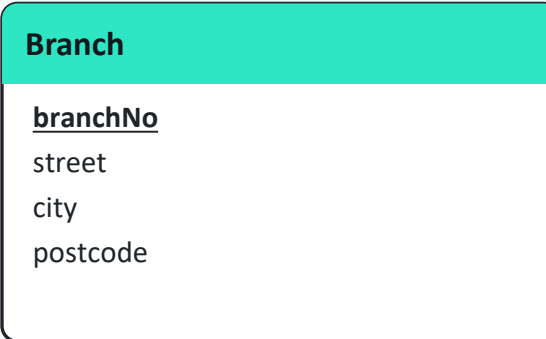
COMSATS University Islamabad, Lahore Campus

What This Lecture Covers

- 1 Entity types and entity occurrences
- 2 Relationship types and relationship occurrences
- 3 Five attribute categories: simple, composite, single-valued, multi-valued, derived
- 4 Strong entity types vs. weak entity types
- 5 Attributes that belong to a relationship itself
- 6 Structural constraints: cardinality (1:1, 1:N, M:N) and participation (mandatory/optional)

Entities, Relationships, Attributes

- The ER model represents an organization's data as entities (things worth tracking), relationships (meaningful associations), and attributes (properties of either).
- Introduced by Peter Chen in 1976 to give conceptual modeling a precise, drawable notation both business stakeholders and designers could read.



An entity type — the basic building block

Entity Types and Occurrences

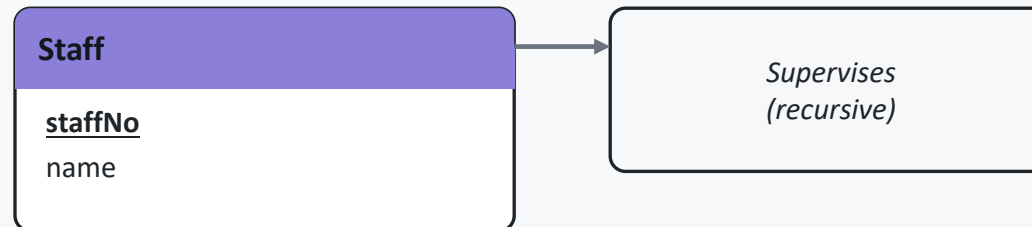
- An entity type is a group of objects with the same properties, significant enough to track independently — Staff, Branch, and PropertyForRent are all entity types.
- An entity occurrence (or entity instance) is one uniquely identifiable member of an entity type — the specific staff member SL21, John White, is one occurrence of the Staff entity type.

NOTE

Entity type vs. relation — same box, different lifecycle. The entity type Branch is a conceptual, technology-free idea; the Branch relation (Lecture 11) is the logical mapping of that idea, complete with a chosen primary key and eventual data types.

Binary, Recursive, and Ternary Relationships

- A relationship type is a meaningful association among entity types. A relationship occurrence is one specific, uniquely identifiable association — “SL21 works at B005” is one occurrence of Has.
- Most relationships in this course are binary — they connect exactly two entity types, like Has connecting Branch and Staff.
- A relationship connecting an entity type to itself is recursive (unary); one connecting three entity types at once is ternary.



“A Staff member Supervises another Staff member” — both participating occurrences come from the same entity type.

Every Attribute Category, In One Entity

PrivateOwner

ownerNo

name

address

telNo {}

numberOfProperties

- Simple (atomic): ownerNo — cannot be meaningfully subdivided.
- Composite: name, address — divisible into meaningful sub-parts.
- Single-valued: ownerNo — exactly one value per occurrence.
- Multi-valued (telNo {}): can hold more than one value per occurrence.
- Derived (numberOfProperties): computed from other data, never stored — storing it would create exactly the redundancy Lecture 11 warned about.

Attribute Categories at a Glance

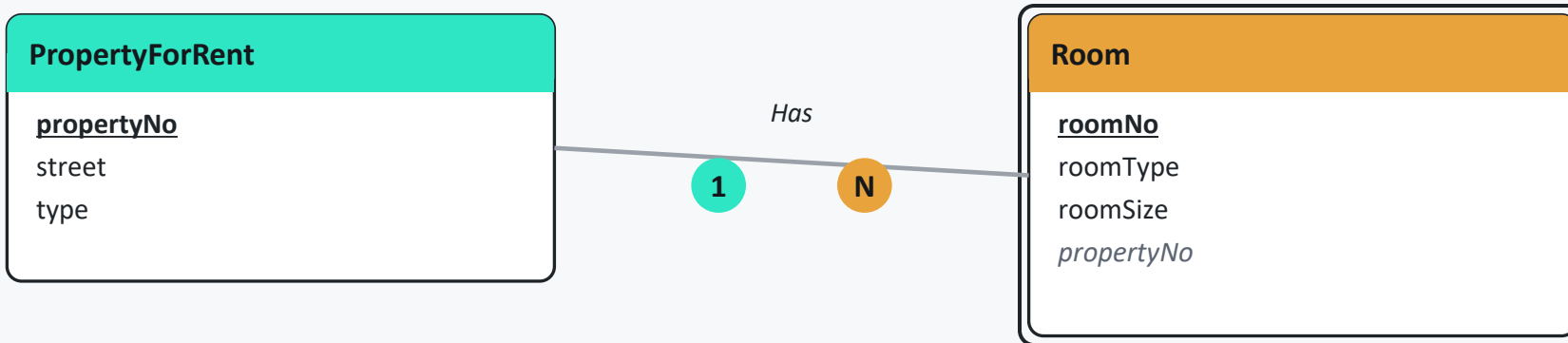
Axis	Category	Meaning
Divisibility	Simple (atomic)	Cannot be meaningfully subdivided further
Divisibility	Composite	Can be divided into meaningful sub-parts
Value count	Single-valued	Holds exactly one value per occurrence
Value count	Multi-valued { }	Can hold more than one value per occurrence
Storage	Derived	Computed from other attributes, never stored directly

WARNING

A multi-valued attribute cannot be stored as one column. Mapping telNo { } to the relational model does not produce one column, but a whole new relation, OwnerTelNo(ownerNo, telNo), with one row per phone number.

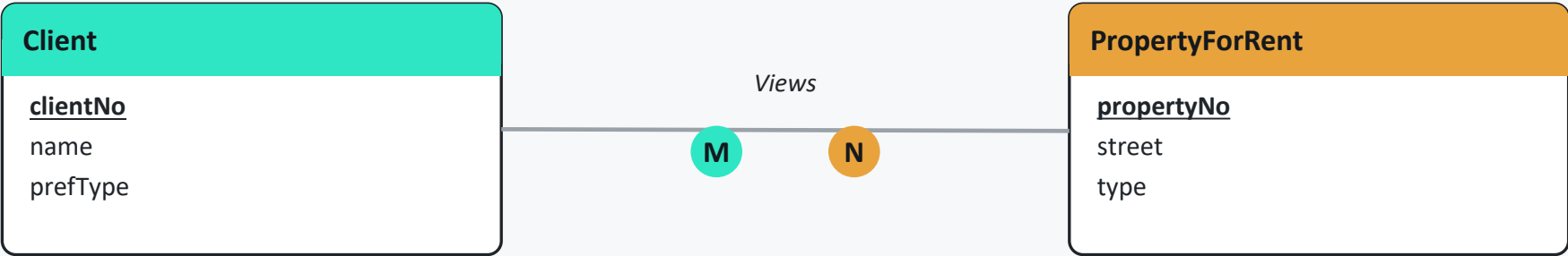
Weak Entity Types

- A strong entity type has independent existence and its own key uniquely identifies every occurrence — Branch, Staff, PropertyForRent, PrivateOwner.
- A weak entity type depends on an owner entity type, and its own key is only a partial key — unique within one owner occurrence, not across the whole entity type.



Room is a weak entity — double border, existence depends on PropertyForRent. True key: (propertyNo, roomNo).

Attributes That Belong to the Pairing



Views carries its own attributes

Attribute of Views	Meaning
viewDate	The date this specific client viewed this specific property
comment	Feedback left after that specific viewing

Neither attribute belongs to Client alone or PropertyForRent alone — each only makes sense attached to one pairing. This is why an M:N relationship with attributes maps to its own relation: `Viewing(clientNo, propertyNo, viewDate, comment)`.

Cardinality (max) + Participation (min)

- Drawing M and N next to a relationship diamond communicates how many occurrences can pair up — but says nothing about whether pairing up is required.
- A complete description needs both pieces, together called structural constraints, usually written as a (min, max) pair for each participating entity type.



Cardinality

The maximum number of relationship occurrences an entity occurrence can participate in.

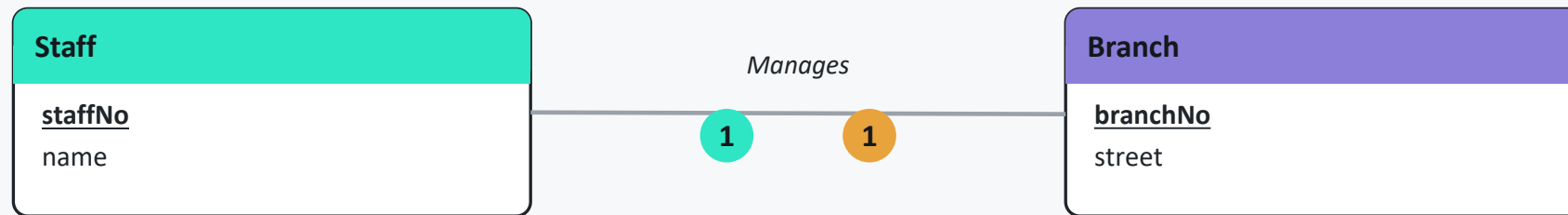


Participation

The minimum number of relationship occurrences an entity occurrence must participate in (0 or 1, almost always).

One-to-One (1:1)

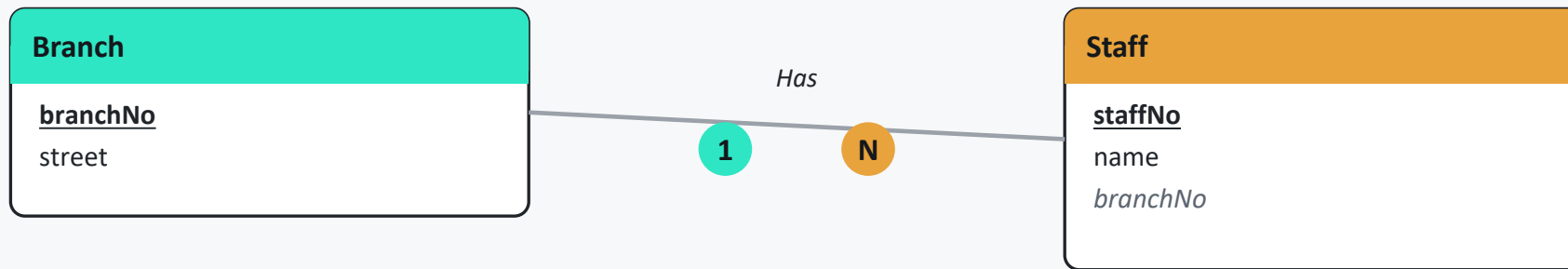
Every occurrence on each side is associated with at most one occurrence on the other side.



A Staff member manages at most one Branch, and a Branch is managed by at most one Staff member.

One-to-Many (1:N)

One occurrence on one side can be associated with many on the other, but each of those many relates back to only one.



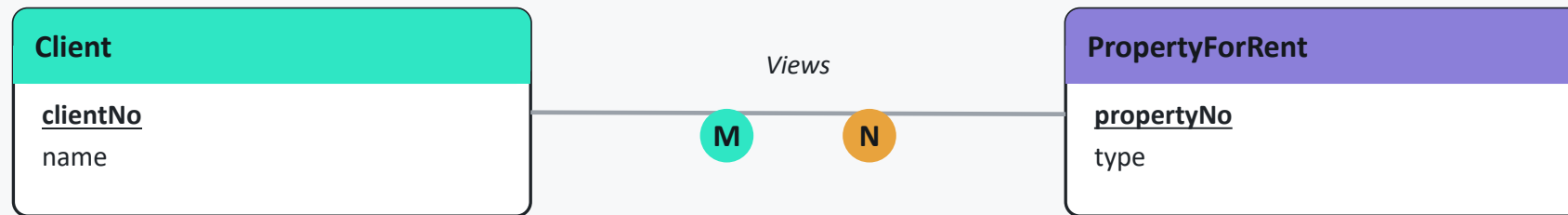
One Branch employs many Staff, but each Staff member works at exactly one Branch.

TIP

Reading cardinality off a diagram: “look across, not down.” The badge next to Staff (N) answers “how many Staff occurrences can one Branch connect to?”

Many-to-Many (M:N)

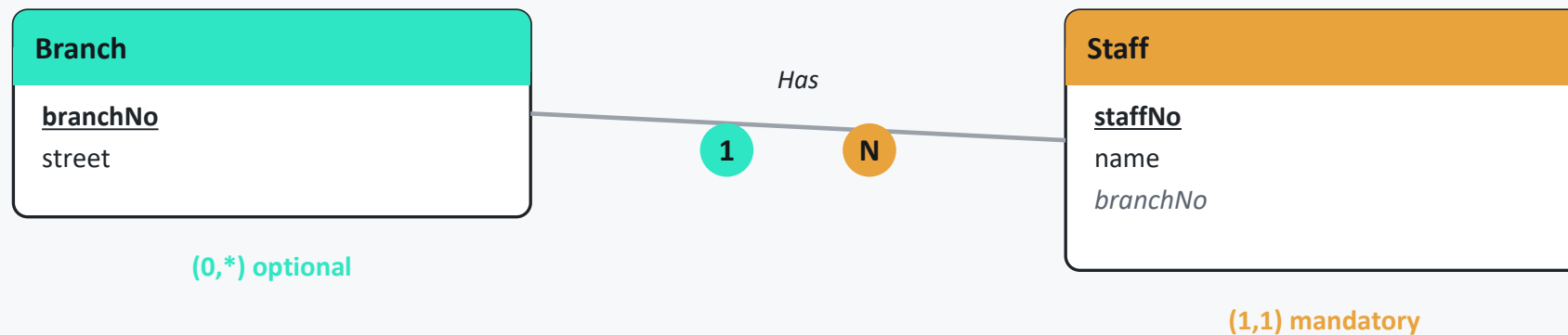
Occurrences on either side can be associated with many occurrences on the other.



Many Clients can view many PropertyForRent listings, and each property can be viewed by many different clients.

Mandatory vs. Optional Participation

- Mandatory (total): every occurrence of the entity type must appear in at least one relationship occurrence; min = 1. Drawn with a double line.
- Optional (partial): an occurrence is allowed to appear in zero relationship occurrences; min = 0. Drawn with a single line.



Staff.branchNo is never NULL — every staff member must work at exactly one branch. A newly opened Branch may legitimately have zero staff assigned yet.

Cardinality and Participation Are Independent

WARNING

It is tempting to read “1:N” and assume the “1” side is automatically mandatory. It is not. Cardinality asks “how many can there be?” (the 1/N/M badges); participation asks “must there be at least one?” (mandatory vs. optional). These are two separate questions about two separate things.

A relationship needs both answered, for each side, to be fully specified — that's exactly why they're both listed together as one (min, max) pair.

KEY TAKEAWAYS

The Entity-Relationship Model

- 1 The ER model represents data as entity types (things), relationship types (associations), and attributes (properties).
- 2 Attributes divide along two axes plus one extra: simple vs. composite, single-valued vs. multi-valued, and derived (computed, never stored).
- 3 A strong entity type has independent existence and a key that identifies it alone; a weak entity type depends on an owner entity and has only a partial key.
- 4 A relationship type can carry its own attributes when a fact belongs to the pairing, not to either participant — mapping it to its own relation.
- 5 Structural constraints combine cardinality (the maximum: 1:1, 1:N, M:N) and participation (the minimum: mandatory or optional) — two genuinely independent questions.